Epoxy Adhesives

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Two-Part Epoxy Adhesive	IS 9197	ISO 4587 (Tensile Shear Strength of Adhesive Bonds); ISO 11343 (Dynamic Impact Wedge-Peel Strength); ISO 10365 (Adhesive Failure Classification); ISO 527 (Mechanical Properties where applicable),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D1876 (T-Peel Strength); ASTM D905 (Compression Shear Strength); ASTM D2094 (Tensile Properties of Adhesive Bonds),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN 1465 (Tensile Lap-Shear Strength); DIN EN ISO 11339 (T-Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Modes); DIN EN ISO 527 (Mechanical Properties); DIN EN ISO 604 (Compression Properties where applicable)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, reactive diluents, curing agents, SVHC substances); REACH Annex XVII (epoxy sensitizers where applicable); FDA: 21 CFR 175.105 where used in food-contact adhesive applications
One-Part Epoxy Adhesive	IS 9197	ISO 4587 (Tensile Shear Strength); ISO 10365 (Failure Classification); ISO 527 (Mechanical Properties),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D905 (Compression Shear Strength); ASTM D1876 (Peel Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Classification)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, curing agents, additives); FDA: 21 CFR 175.105 where food-contact compliant formulation is used
Structural Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 11343 (Impact Peel Strength); ISO 10365 (Failure Classification); ISO 527 (Mechanical Properties),ISO 19095	ASTM D3165 (Strength Properties of Adhesives in Shear by Tension Loading); ASTM D1002 (Lap Shear Strength); ASTM D1876 (T-Peel Strength); ASTM D3762 (Aluminium Adhesive Bond Strength); ASTM D1183 (Durability of Adhesive-Bonded Structures),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Environmental Durability); DIN EN ISO 10365 (Failure Pattern); DIN EN ISO 527 (Mechanical Strength)	REACH: Regulation (EC) No. 1907/2006 (epoxy resin systems, hardeners, additives); AMPP: SP 1/SP 6 where bonding involves prepared metal substrates
Electrical Grade Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 527 (Mechanical Properties); ISO 14644 (Cleanroom Applications where applicable); ISO electrical insulation material testing standards where applicable,ISO 19095	ASTM D149 (Dielectric Breakdown Voltage); ASTM D257 (Electrical Resistance); ASTM D1002 (Shear Strength); ASTM D3165 (Adhesive Strength); ASTM D2307 (Thermal Endurance of Electrical Insulation Materials where applicable),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN IEC 60243 series (Electric Strength); DIN EN IEC 60093 (Volume Resistivity); DIN EN IEC 62631 series (Dielectric Properties); DIN EN 1465 (Mechanical Strength); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, flame retardants, fillers, additives); RoHS Directive 2011/65/EU where used in electrical/electronic assemblies
Thermally Conductive Epoxy Adhesive	IS 9197	ISO 22007 Series (Thermal Conductivity of Plastics); ISO 4587 (Bond Strength); ISO 527 (Mechanical Properties),ISO 19095	ASTM E1461 (Thermal Diffusivity by Flash Method); ASTM D5470 (Thermal Transmission Properties of Thermal Interface Materials); ASTM D1002 (Shear Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 22007 series (Thermal Conductivity of Plastics); DIN EN 1465 (Bond Strength); DIN EN IEC 62631 (Electrical Properties where applicable); DIN EN ISO 527 (Mechanical Properties)	REACH: Regulation (EC) No. 1907/2006 (epoxy resin, aluminium/boron nitride fillers, additives); RoHS Directive 2011/65/EU where used in electronic applications
Electrically Conductive Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO electrical resistivity testing methods; ISO 527 (Mechanical Properties),ISO 19095	ASTM B193 (Electrical Resistivity of Conductive Materials); ASTM D257 (Electrical Resistance); ASTM D1002 (Shear Strength); ASTM D149 (Dielectric Breakdown where insulation properties apply),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN IEC 60093 (Volume Resistivity); DIN EN IEC 62631 series (Dielectric Properties); DIN EN IEC 60243 (Electric Strength); DIN EN 1465 (Mechanical Bond Strength)	REACH: Regulation (EC) No. 1907/2006 (silver, carbon, metal fillers, epoxy chemistry); RoHS Directive 2011/65/EU where used in electronic assemblies
Metal Bonding Epoxy	IS 9197	ISO 4587 (Tensile Shear Strength);	ASTM D1002 (Lap Shear Strength);	DIN EN 1465 (Lap Shear Strength);	AMPP: SP 1 (Solvent Cleaning); AMPP:

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		ISO 12944 (Corrosion Protection where applicable); ISO 10365 (Failure Classification),ISO 19095	ASTM D3762 (Adhesive Bond Strength of Aluminium); ASTM D3165 (Shear Strength); ASTM D903 (Peel Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Classification)	SP 2/SP 3 where surface preparation is required; REACH: Regulation (EC) No. 1907/2006 (epoxy resins, curing agents, additives)
Concrete Bonding Epoxy	IS 9197	ISO 4624 (Pull-Off Adhesion); ISO 4587 (Bond Strength); ISO 10365 (Failure Classification),ISO 19095	ASTM C881 (Epoxy-Resin-Base Bonding Systems for Concrete); ASTM D4541 (Pull-Off Adhesion); ASTM C1583 (Tensile Strength of Concrete Surfaces); ASTM D1002 (Bond Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN 1504-4 (Structural Bonding for Concrete); DIN EN 1542 (Products and Systems for Protection and Repair of Concrete — Adhesion); DIN EN 1465 (Bond Strength); DIN EN ISO 9142 (Durability)	AMPP: SP 13/NACE No.6 (Surface Preparation of Concrete); REACH: Regulation (EC) No. 1907/2006 (epoxy resins, curing agents, additives)
Ceramic Bonding Epoxy	IS 9197	ISO 4587 (Bond Strength); ISO 10365 (Failure Classification); ISO 527 (Mechanical Properties),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D1876 (T-Peel Strength); ASTM D905 (Compression Shear Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Mode)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, fillers, additives); FDA: only where used in regulated food-contact ceramic applications
Glass Bonding Epoxy	IS 9197	ISO 4587 (Bond Strength); ISO 8510 (Peel Adhesion); ISO 10365 (Failure Classification),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D903 (Peel Strength); ASTM D3165 (Shear Strength); ASTM D3359 (Adhesion),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Classification)	REACH: Regulation (EC) No. 1907/2006 (epoxy components, additives); FDA: 21 CFR 175.105 where food-contact glass applications exist
Wood Bonding Epoxy	IS 9197	ISO 12465 (Wood Adhesive Applications where applicable); ISO 4587 (Bond Strength); ISO 11339 (Peel Strength),ISO 19095	ASTM D905 (Wood Adhesive Strength); ASTM D5751 (Adhesives Used in Structural Wood Products); ASTM D1002 (Shear Strength); ASTM D1876 (Peel Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN 205 (Wood Adhesive Shear Strength); DIN EN 204 (Classification of Wood Adhesives); DIN EN 14257 (WATT 91 Heat Resistance); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, hardeners); FDA: 21 CFR 175.105 only where food-contact wood applications exist
Marine Grade Epoxy Adhesive	IS 9197	ISO 12944 (Corrosion Protection); ISO 9227 (Salt Spray Testing); ISO 4587 (Bond Strength),ISO 19095	ASTM D1183 (Durability of Adhesive Bonds); ASTM B117 (Salt Spray where corrosion protection applies); ASTM D1002 (Shear Strength); ASTM D903 (Peel Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 12944 series (Corrosion Protection where applicable); DIN EN ISO 9227 (Salt Spray); DIN EN ISO 9142 (Durability); DIN EN 1465 (Shear Strength)	AMPP: SP 1; AMPP: SP 6/NACE No.3 where bonding coated/metal marine structures; REACH: Regulation (EC) No. 1907/2006 (marine-use chemicals, solvents, additives)
Aerospace Grade Epoxy Adhesive	IS 9197	ISO 9100 (Aerospace Quality Requirements where applicable); ISO 4587 (Bond Strength); ISO 11343 (Impact Peel Strength); ISO 527 (Mechanical Properties),ISO 19095	ASTM D3165 (Shear Strength); ASTM D1002 (Lap Shear Strength); ASTM D1876 (Peel Strength); ASTM D2294 (Bond Strength after Environmental Exposure),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Environmental Durability); DIN EN ISO 10365 (Failure Classification)	REACH: Regulation (EC) No. 1907/2006 (epoxy chemistry, curing agents, SVHC substances); RoHS where electrical aerospace components are involved
Automotive Epoxy Adhesive	IS 9197	ISO 16750 (Road Vehicle Environmental Testing); ISO 4587 (Bond Strength); ISO 11343 (Impact Peel Strength),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D3762 (Metal Bonding Strength); ASTM D3165 (Shear Strength); ASTM D1183 (Durability Testing),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 9142 (Durability); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 10365 (Failure Mode)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, VOC, restricted substances); REACH Annex XVII where applicable

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Chemical Resistant Epoxy Adhesive	IS 9197	ISO 2812 Series (Resistance to Liquids); ISO 4587 (Bond Strength); ISO 10365 (Failure Classification),ISO 19095	ASTM D543 (Chemical Resistance of Plastics); ASTM D1308 (Chemical Resistance of Coatings); ASTM D1002 (Shear Strength); ASTM D903 (Peel Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 2812 series (Resistance to Liquids); DIN EN ISO 9142 (Chemical Durability); DIN EN 1465 (Shear Strength); DIN EN ISO 11339 (Peel Strength)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, hardeners, chemical-resistant additives); AMPP: SP 1/SP 6 where applied on metal substrates
High Temperature Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 527 (Mechanical Properties); ISO 10365 (Failure Classification),ISO 19095	ASTM D2294 (Adhesive Bond Strength after Heat Exposure); ASTM D1002 (Shear Strength); ASTM D3165 (Shear Strength); ASTM D905 (Compression Shear Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN ISO 9142 (Temperature Durability); DIN EN 1465 (Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 10365 (Failure Classification)	REACH: Regulation (EC) No. 1907/2006 (high-temperature epoxy systems, fillers, additives)
Gap Filling Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 10365 (Failure Classification); ISO 527 (Mechanical Properties),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D905 (Compression Shear Strength); ASTM D3165 (Shear Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 604 (Compression Strength); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, fillers, curing agents); RoHS where used in electronic assemblies
Fast Cure Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 10365 (Failure Classification),ISO 19095	ASTM D5895 (Drying/Curing Time); ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 1465 (Lap Shear Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 10365 (Failure Pattern)	REACH: Regulation (EC) No. 1907/2006 (reactive diluents, accelerators, curing agents)
Clear Epoxy Adhesive	IS 9197	ISO 4587 (Bond Strength); ISO 2813 (Gloss where optical finish applies); ISO 10365 (Failure Classification),ISO 19095	ASTM D1002 (Lap Shear Strength); ASTM D523 (Gloss where optical clarity applies); ASTM D2244 (Colour Difference); ASTM D3359 (Adhesion),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 1465 (Bond Strength); DIN EN ISO 2813 (Gloss); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (epoxy resins, reactive diluents, additives); FDA: 21 CFR 175.105 where food-contact applications exist
Medical Grade Epoxy Adhesive	IS 9197	ISO 10993 Series (Biological Evaluation of Medical Devices); ISO 13485 (Medical Device Adhesive Application Quality Requirements where applicable); ISO 4587,ISO 19095	ASTM F88 (Seal Strength where medical packaging applies); ASTM D1002 (Shear Strength); ASTM F2026 (Medical Device Adhesive Applications where applicable); ASTM D543 (Chemical Resistance),ASTM D1184; ASTM D2093; ASTM D3163; ASTM D3529; ASTM D3807; ASTM D905	DIN EN ISO 10993 series (Biological Evaluation of Medical Devices where applicable); DIN EN 1465 (Bond Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 2812 (Chemical Resistance)	REACH: Regulation (EC) No. 1907/2006; FDA: applicable medical device adhesive requirements where used in FDA-regulated devices; ISO 10993 series where biological evaluation is required

Source field: Dyes and Pigments